

LAST NAME: \_\_\_\_\_

## Practice Quiz 07C — MATH 2055 — Fall 2026

Practice for Quiz 07

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

*Revised using GPT-6 Astra in the Codex harness.*

1. Let  $f : \mathbf{R} \rightarrow \mathbf{R}$  and let  $c, L \in \mathbf{R}$ . Suppose  $\lim_{x \rightarrow c} f(x) = L$ . Prove directly from the epsilon-delta definition that

$$\lim_{x \rightarrow c} (-3f(x)) = -3L.$$